

## High-Speed CMOS Logic 8-Bit Shift Register with Input Storage

### Features

- Buffered Inputs
- Asynchronous Parallel Load
- Fanout (Over Temperature Range)
  - Standard Outputs . . . . . 10 LSTTL Loads
  - Bus Driver Outputs . . . . . 15 LSTTL Loads
- Wide Operating Temperature Range . . . -55°C to 125°C
- Balanced Propagation Delay and Transition Times
- Significant Power Reduction Compared to LSTTL Logic ICs
- HC Types
  - 2V to 6V Operation
  - High Noise Immunity:  $N_{IL} = 30\%$ ,  $N_{IH} = 30\%$  of  $V_{CC}$  at  $V_{CC} = 5V$
- HCT Types
  - 4.5V to 5.5V Operation
  - Direct LSTTL Input Logic Compatibility,  $V_{IL} = 0.8V$  (Max),  $V_{IH} = 2V$  (Min)
  - CMOS Input Compatibility,  $I_I \leq 1\mu A$  at  $V_{OL}$ ,  $V_{OH}$

### Description

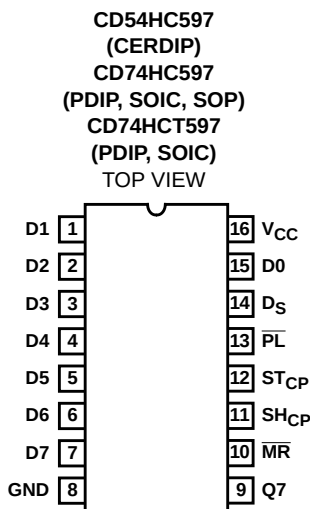
The 'HC597 and CD74HCT597 are high-speed silicon gate CMOS devices that are pin-compatible with the LSTTL 597 devices. Each device consists of an 8-flip-flop input register and an 8-bit parallel-in/serial-in, serial-out shift register. Each register is controlled by its own clock. A "low" on the parallel load input ( $\overline{PL}$ ) shifts parallel stored data asynchronously into the shift register. A "low" master input ( $\overline{MR}$ ) clears the shift register. Serial input data can also be synchronously shifted through the shift register when  $\overline{PL}$  is high.

### Ordering Information

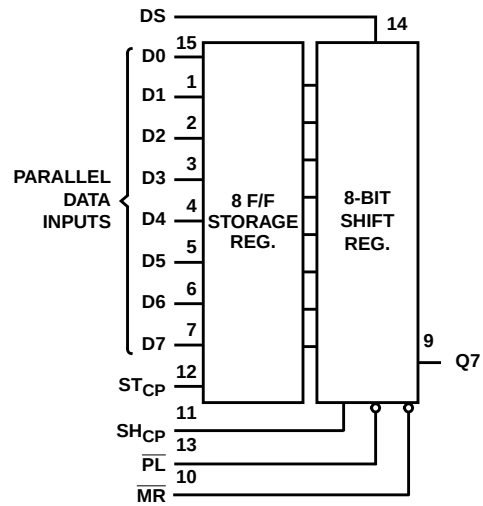
| PART NUMBER   | TEMP. RANGE (°C) | PACKAGE      |
|---------------|------------------|--------------|
| CD54HC597F3A  | -55 to 125       | 16 Ld CERDIP |
| CD74HC597E    | -55 to 125       | 16 Ld PDIP   |
| CD74HC597M    | -55 to 125       | 16 Ld SOIC   |
| CD74HC597MT   | -55 to 125       | 16 Ld SOIC   |
| CD74HC597M96  | -55 to 125       | 16 Ld SOIC   |
| CD74HC597NSR  | -55 to 125       | 16 Ld SOP    |
| CD74HCT597E   | -55 to 125       | 16 Ld PDIP   |
| CD74HCT597M   | -55 to 125       | 16 Ld SOIC   |
| CD74HCT597MT  | -55 to 125       | 16 Ld SOIC   |
| CD74HCT597M96 | -55 to 125       | 16 Ld SOIC   |

NOTE: When ordering, use the entire part number. The suffixes 96 and R denote tape and reel. The suffix T denotes a small-quantity reel of 250.

### Pinout



## Functional Diagram



FUNCTION TABLE

| ST <sub>CP</sub> | SH <sub>CP</sub> | $\overline{PL}$ | $\overline{MR}$ | FUNCTION                                                                  |
|------------------|------------------|-----------------|-----------------|---------------------------------------------------------------------------|
| ↑                | X                | X               | X               | Data Loaded to Input Flip-Flops                                           |
| ↑                | X                | L               | H               | Data Loaded from Inputs to Shift Register                                 |
| No Clock Edge    | X                | L               | H               | Data Transferred from Input Flip-Flops to Shift Register                  |
| X                | X                | L               | L               | Invalid Logic, State of Shift Register Indeterminate when Signals Removed |
| X                | X                | H               | L               | Shift Register Cleared                                                    |
| X                | ↑                | H               | H               | Shift Register Clocked $Q_n = Q_{n-1}$ , $Q_0 = D_S$                      |

H = High Voltage Level, L = Low Voltage Level, X = Don't Care, ↑ = Transition from Low to High CP Level

# CD54HC597, CD74HC597, CD74HCT597

## Absolute Maximum Ratings

|                                                        |             |
|--------------------------------------------------------|-------------|
| DC Supply Voltage, $V_{CC}$                            | -0.5V to 7V |
| DC Input Diode Current, $I_{IK}$                       |             |
| For $V_I < -0.5V$ or $V_I > V_{CC} + 0.5V$             | $\pm 20mA$  |
| DC Output Diode Current, $I_{OK}$                      |             |
| For $V_O < -0.5V$ or $V_O > V_{CC} + 0.5V$             | $\pm 20mA$  |
| DC Drain Current, per Output, $I_O$                    |             |
| For $-0.5V < V_O < V_{CC} + 0.5V$                      | $\pm 25mA$  |
| DC Output Source or Sink Current per Output Pin, $I_O$ |             |
| For $V_O > -0.5V$ or $V_O < V_{CC} + 0.5V$             | $\pm 25mA$  |
| DC $V_{CC}$ or Ground Current, $I_{CC}$                | $\pm 50mA$  |

## Thermal Information

|                                          |                                    |
|------------------------------------------|------------------------------------|
| Thermal Resistance (Typical, Note 1)     | $\theta_{JA}$ ( $^{\circ}C/W$ )    |
| E (PDIP) Package                         | 67                                 |
| M (SOIC) Package                         | 73                                 |
| NS (SOP) Package                         | 64                                 |
| Maximum Junction Temperature             | 150 $^{\circ}C$                    |
| Maximum Storage Temperature Range        | -65 $^{\circ}C$ to 150 $^{\circ}C$ |
| Maximum Lead Temperature (Soldering 10s) | 300 $^{\circ}C$                    |
| (SOIC - Lead Tips Only)                  |                                    |

## Operating Conditions

|                                           |                                    |
|-------------------------------------------|------------------------------------|
| Temperature Range, $T_A$                  | -55 $^{\circ}C$ to 125 $^{\circ}C$ |
| Supply Voltage Range, $V_{CC}$            |                                    |
| HC Types                                  | .2V to 6V                          |
| DC Input or Output Voltage, $V_I$ , $V_O$ | 0V to $V_{CC}$                     |
| Input Rise and Fall Time                  |                                    |
| 2V                                        | 1000ns (Max)                       |
| 4.5V                                      | 500ns (Max)                        |
| 6V                                        | 400ns (Max)                        |

**CAUTION:** Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

### NOTE:

1. The package thermal impedance is calculated in accordance with JESD 51-7.

## DC Electrical Specifications

| PARAMETER                               | SYMBOL          | TEST CONDITIONS                    |                     | V <sub>CC</sub> (V) | 25°C |      |      | -40°C TO 85°C |      | -55°C TO 125°C |      | UNITS |
|-----------------------------------------|-----------------|------------------------------------|---------------------|---------------------|------|------|------|---------------|------|----------------|------|-------|
|                                         |                 | V <sub>I</sub> (V)                 | I <sub>O</sub> (mA) |                     | MIN  | TYP  | MAX  | MIN           | MAX  | MIN            | MAX  |       |
| HC TYPES                                |                 |                                    |                     |                     |      |      |      |               |      |                |      |       |
| High Level Input Voltage                | V <sub>IH</sub> | -                                  | -                   | 2                   | 1.5  | -    | -    | 1.5           | -    | 1.5            | -    | V     |
|                                         |                 |                                    |                     | 4.5                 | 3.15 | -    | -    | 3.15          | -    | 3.15           | -    | V     |
|                                         |                 |                                    |                     | 6                   | 4.2  | -    | -    | 4.2           | -    | 4.2            | -    | V     |
| Low Level Input Voltage                 | V <sub>IL</sub> | -                                  | -                   | 2                   | -    | -    | 0.5  | -             | 0.5  | -              | 0.5  | V     |
|                                         |                 |                                    |                     | 4.5                 | -    | -    | 1.35 | -             | 1.35 | -              | 1.35 | V     |
|                                         |                 |                                    |                     | 6                   | -    | -    | 1.8  | -             | 1.8  | -              | 1.8  | V     |
| High Level Output Voltage<br>CMOS Loads | V <sub>OH</sub> | V <sub>IH</sub> or V <sub>IL</sub> | -0.02               | 2                   | 1.9  | -    | -    | 1.9           | -    | 1.9            | -    | V     |
|                                         |                 |                                    | -0.02               | 4.5                 | 4.4  | -    | -    | 4.4           | -    | 4.4            | -    | V     |
|                                         |                 |                                    | -0.02               | 6                   | 5.9  | -    | -    | 5.9           | -    | 5.9            | -    | V     |
| -                                       |                 |                                    | -                   | -                   | -    | -    | -    | -             | -    | -              | V    |       |
| -4                                      |                 |                                    | 4.5                 | 3.98                | -    | -    | 3.84 | -             | 3.7  | -              | V    |       |
| -5.2                                    |                 |                                    | 6                   | 5.48                | -    | -    | 5.34 | -             | 5.2  | -              | V    |       |
| High Level Output Voltage<br>TTL Loads  | V <sub>OL</sub> | V <sub>IH</sub> or V <sub>IL</sub> | -                   | -                   | -    | -    | -    | -             | -    | -              | -    | V     |
|                                         |                 |                                    | -                   | -                   | -    | -    | -    | -             | -    | -              | -    | V     |
|                                         |                 |                                    | -                   | -                   | -    | -    | -    | -             | -    | -              | -    | V     |
| 0.02                                    |                 |                                    | 2                   | -                   | -    | 0.1  | -    | 0.1           | -    | 0.1            | V    |       |
| 0.02                                    |                 |                                    | 4.5                 | -                   | -    | 0.1  | -    | 0.1           | -    | 0.1            | V    |       |
| 0.02                                    |                 |                                    | 6                   | -                   | -    | 0.1  | -    | 0.1           | -    | 0.1            | V    |       |
| Low Level Output Voltage<br>CMOS Loads  | V <sub>OL</sub> | V <sub>IH</sub> or V <sub>IL</sub> | -                   | -                   | -    | -    | -    | -             | -    | -              | -    | V     |
|                                         |                 |                                    | -                   | -                   | -    | -    | -    | -             | -    | -              | -    | V     |
|                                         |                 |                                    | -                   | -                   | -    | -    | -    | -             | -    | -              | -    | V     |
| 4                                       |                 |                                    | 4.5                 | -                   | -    | 0.26 | -    | 0.33          | -    | 0.4            | V    |       |
| 5.2                                     |                 |                                    | 6                   | -                   | -    | 0.26 | -    | 0.33          | -    | 0.4            | V    |       |
| 5.2                                     |                 |                                    | 6                   | -                   | -    | 0.26 | -    | 0.33          | -    | 0.4            | V    |       |
| Low Level Output Voltage<br>TTL Loads   | V <sub>OL</sub> | V <sub>IH</sub> or V <sub>IL</sub> | -                   | -                   | -    | -    | -    | -             | -    | -              | V    |       |
| -                                       |                 |                                    | -                   | -                   | -    | -    | -    | -             | -    | V              |      |       |
| -                                       |                 |                                    | -                   | -                   | -    | -    | -    | -             | -    | V              |      |       |
| Input Leakage Current                   | I <sub>I</sub>  | V <sub>CC</sub> or GND             | -                   | 6                   | -    | -    | ±0.1 | -             | ±1   | -              | ±1   | μA    |

# CD54HC597, CD74HC597, CD74HCT597

## DC Electrical Specifications (Continued)

| PARAMETER                                                      | SYMBOL                   | TEST CONDITIONS      |            | $V_{CC}$ (V) | 25°C |     |           | -40°C TO 85°C |         | -55°C TO 125°C |         | UNITS   |
|----------------------------------------------------------------|--------------------------|----------------------|------------|--------------|------|-----|-----------|---------------|---------|----------------|---------|---------|
|                                                                |                          | $V_I$ (V)            | $I_O$ (mA) |              | MIN  | TYP | MAX       | MIN           | MAX     | MIN            | MAX     |         |
| Quiescent Device Current                                       | $I_{CC}$                 | $V_{CC}$ or GND      | 0          | 6            | -    | -   | 8         | -             | 80      | -              | 160     | $\mu A$ |
| <b>HCT TYPES</b>                                               |                          |                      |            |              |      |     |           |               |         |                |         |         |
| High Level Input Voltage                                       | $V_{IH}$                 | -                    | -          | 4.5 to 5.5   | 2    | -   | -         | 2             | -       | 2              | -       | V       |
| Low Level Input Voltage                                        | $V_{IL}$                 | -                    | -          | 4.5 to 5.5   | -    | -   | 0.8       | -             | 0.8     | -              | 0.8     | V       |
| High Level Output Voltage<br>CMOS Loads                        | $V_{OH}$                 | $V_{IH}$ or $V_{IL}$ | -0.02      | 4.5          | 4.4  | -   | -         | 4.4           | -       | 4.4            | -       | V       |
| High Level Output Voltage<br>TTL Loads                         |                          |                      | -4         | 4.5          | 3.98 | -   | -         | 3.84          | -       | 3.7            | -       | V       |
| Low Level Output Voltage<br>CMOS Loads                         | $V_{OL}$                 | $V_{IH}$ or $V_{IL}$ | 0.02       | 4.5          | -    | -   | 0.1       | -             | 0.1     | -              | 0.1     | V       |
| Low Level Output Voltage<br>TTL Loads                          |                          |                      | 4          | 4.5          | -    | -   | 0.26      | -             | 0.33    | -              | 0.4     | V       |
| Input Leakage Current                                          | $I_I$                    | $V_{CC}$ and GND     | 0          | 5.5          | -    |     | $\pm 0.1$ | -             | $\pm 1$ | -              | $\pm 1$ | $\mu A$ |
| Quiescent Device Current                                       | $I_{CC}$                 | $V_{CC}$ or GND      | 0          | 5.5          | -    | -   | 8         | -             | 80      | -              | 160     | $\mu A$ |
| Additional Quiescent Device Current Per Input Pin: 1 Unit Load | $\Delta I_{CC}$ (Note 2) | $V_{CC}$ -2.1        | -          | 4.5 to 5.5   | -    | 100 | 360       | -             | 450     | -              | 490     | $\mu A$ |

NOTE:

- For dual-supply systems theoretical worst case ( $V_I = 2.4V$ ,  $V_{CC} = 5.5V$ ) specification is 1.8mA.

## HCT Input Loading Table

| INPUT                 | UNIT LOADS |
|-----------------------|------------|
| $D_S$                 | 0.2        |
| $D_n$                 | 0.3        |
| $\overline{PL}$ , MR  | 1.5        |
| $ST_{CP}$ , $SH_{CP}$ | 1.5        |

NOTE: Unit Load is  $\Delta I_{CC}$  limit specified in DC Electrical Specifications Table, e.g., 360 $\mu A$  max. at 25°C.

## Prerequisite for Switching Specifications

| PARAMETER                  | SYMBOL           | V <sub>CC</sub> (V) | 25°C |     |     | -40°C TO 85°C |     |     | -55°C TO 125°C |     |     | UNITS |
|----------------------------|------------------|---------------------|------|-----|-----|---------------|-----|-----|----------------|-----|-----|-------|
|                            |                  |                     | MIN  | TYP | MAX | MIN           | TYP | MAX | MIN            | TYP | MAX |       |
| HC TYPES                   |                  |                     |      |     |     |               |     |     |                |     |     |       |
| SH <sub>CP</sub> Frequency | f <sub>MAX</sub> | 2                   | 6    | -   | -   | 5             | -   | -   | 4              | -   | -   | MHz   |
|                            |                  | 4.5                 | 30   | -   | -   | 25            | -   | -   | 20             | -   | -   | MHz   |
|                            |                  | 6                   | 35   | -   | -   | 29            | -   | -   | 23             | -   | -   | MHz   |

**CD54HC597, CD74HC597, CD74HCT597**

**Prerequisite for Switching Specifications (Continued)**

| PARAMETER                                                                                      | SYMBOL           | V <sub>CC</sub> (V) | 25°C |     |     | -40°C TO 85°C |     |     | -55°C TO 125°C |     |     | UNITS |
|------------------------------------------------------------------------------------------------|------------------|---------------------|------|-----|-----|---------------|-----|-----|----------------|-----|-----|-------|
|                                                                                                |                  |                     | MIN  | TYP | MAX | MIN           | TYP | MAX | MIN            | TYP | MAX |       |
| SH <sub>CP</sub> Pulse Width                                                                   | t <sub>W</sub>   | 2                   | 80   | -   | -   | 100           | -   | -   | 120            | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 16   | -   | -   | 20            | -   | -   | 24             | -   | -   | ns    |
|                                                                                                |                  | 6                   | 14   | -   | -   | 17            | -   | -   | 20             | -   | -   | ns    |
| ST <sub>CP</sub> Pulse Width                                                                   | t <sub>W</sub>   | 2                   | 60   | -   | -   | 75            | -   | -   | 90             | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 12   | -   | -   | 15            | -   | -   | 18             | -   | -   | ns    |
|                                                                                                |                  | 6                   | 10   | -   | -   | 13            | -   | -   | 15             | -   | -   | ns    |
| MR Pulse Width                                                                                 | t <sub>W</sub>   | 2                   | 80   | -   | -   | 100           | -   | -   | 120            | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 16   | -   | -   | 20            | -   | -   | 24             | -   | -   | ns    |
|                                                                                                |                  | 6                   | 14   | -   | -   | 17            | -   | -   | 20             | -   | -   | ns    |
| PL Pulse Width                                                                                 | t <sub>W</sub>   | 2                   | 70   | -   | -   | 90            | -   | -   | 105            | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 14   | -   | -   | 18            | -   | -   | 21             | -   | -   | ns    |
|                                                                                                |                  | 6                   | 12   | -   | -   | 15            | -   | -   | 18             | -   | -   | ns    |
| ST <sub>CP</sub> to SH <sub>CP</sub> Setup Time                                                | t <sub>SU</sub>  | 2                   | 100  | -   | -   | 125           | -   | -   | 150            | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 20   | -   | -   | 25            | -   | -   | 30             | -   | -   | ns    |
|                                                                                                |                  | 6                   | 17   | -   | -   | 21            | -   | -   | 26             | -   | -   | ns    |
| D <sub>S</sub> to SH <sub>CP</sub> Setup Time<br>D <sub>n</sub> to ST <sub>CP</sub> Setup Time | t <sub>SU</sub>  | 2                   | 50   | -   | -   | 65            | -   | -   | 75             | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 10   | -   | -   | 13            | -   | -   | 15             | -   | -   | ns    |
|                                                                                                |                  | 6                   | 9    | -   | -   | 11            | -   | -   | 13             | -   | -   | ns    |
| ST <sub>CP</sub> to SH <sub>CP</sub> Setup Time                                                | t <sub>H</sub>   | 2                   | 0    | -   | -   | 0             | -   | -   | 0              | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 0    | -   | -   | 0             | -   | -   | 0              | -   | -   | ns    |
|                                                                                                |                  | 6                   | 0    | -   | -   | 0             | -   | -   | 0              | -   | -   | ns    |
| D <sub>S</sub> to SH <sub>CP</sub> Hold Time<br>D <sub>n</sub> to ST <sub>CP</sub> Hold Time   | t <sub>H</sub>   | 2                   | 3    | -   | -   | 3             | -   | -   | 3              | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 3    | -   | -   | 3             | -   | -   | 3              | -   | -   | ns    |
|                                                                                                |                  | 6                   | 3    | -   | -   | 3             | -   | -   | 3              | -   | -   | ns    |
| MR to SH <sub>CP</sub> Removal Time                                                            | t <sub>REM</sub> | 2                   | 3    | -   | -   | 3             | -   | -   | 3              | -   | -   | ns    |
|                                                                                                |                  | 4.5                 | 3    | -   | -   | 3             | -   | -   | 3              | -   | -   | ns    |
|                                                                                                |                  | 6                   | 3    | -   | -   | 3             | -   | -   | 3              | -   | -   | ns    |
| HCT TYPES                                                                                      |                  |                     |      |     |     |               |     |     |                |     |     |       |
| SH <sub>CP</sub> Frequency                                                                     | f <sub>MAX</sub> | 4.5                 | 25   | -   | -   | 20            | -   | -   | 16             | -   | -   | MHz   |
| SH <sub>CP</sub> Pulse Width                                                                   | t <sub>W</sub>   | 4.5                 | 20   | -   | -   | 25            | -   | -   | 30             | -   | -   | ns    |
| ST <sub>CP</sub> Pulse Width                                                                   | t <sub>W</sub>   | 4.5                 | 13   | -   | -   | 16            | -   | -   | 20             | -   | -   | ns    |
| MR Pulse Width                                                                                 | t <sub>W</sub>   | 4.5                 | 18   | -   | -   | 23            | -   | -   | 27             | -   | -   | ns    |
| PL Pulse Width                                                                                 | t <sub>W</sub>   | 4.5                 | 16   | -   | -   | 20            | -   | -   | 24             | -   | -   | ns    |
| ST <sub>CP</sub> to SH <sub>CP</sub> Setup Time                                                | t <sub>SU</sub>  | 4.5                 | 24   | -   | -   | 30            | -   | -   | 36             | -   | -   | ns    |

# CD54HC597, CD74HC597, CD74HCT597

## Prerequisite for Switching Specifications (Continued)

| PARAMETER                                                                                      | SYMBOL           | V <sub>CC</sub> (V) | 25°C |     |     | -40°C TO 85°C |     |     | -55°C TO 125°C |     |     | UNITS |
|------------------------------------------------------------------------------------------------|------------------|---------------------|------|-----|-----|---------------|-----|-----|----------------|-----|-----|-------|
|                                                                                                |                  |                     | MIN  | TYP | MAX | MIN           | TYP | MAX | MIN            | TYP | MAX |       |
| D <sub>S</sub> to SH <sub>CP</sub> Setup Time<br>D <sub>N</sub> to ST <sub>CP</sub> Setup Time | t <sub>H</sub>   | 4.5                 | 10   | -   | -   | 13            | -   | -   | 15             | -   | -   | ns    |
| ST <sub>CP</sub> to SH <sub>CP</sub> Hold Time                                                 | t <sub>H</sub>   | 4.5                 | 0    | -   | -   | 0             | -   | -   | 0              | -   | -   | ns    |
| D <sub>S</sub> to SH <sub>CP</sub> Hold Time<br>D <sub>N</sub> to ST <sub>CP</sub> Hold Time   | t <sub>H</sub>   | 4.5                 | 3    | -   | -   | 3             | -   | -   | 3              | -   | -   | ns    |
| MR to SH <sub>CP</sub> Removal Time                                                            | t <sub>REM</sub> | 4.5                 | 10   | -   | -   | 13            | -   | -   | 15             | -   | -   | ns    |

## Switching Specifications Input t<sub>r</sub>, t<sub>f</sub> = 6ns

| PARAMETER                                   | SYMBOL                              | TEST CONDITIONS       | V <sub>CC</sub> (V)                 | 25°C                  |      |     | -40°C to 85°C |     | -55°C to 125°C |     | UNITS |
|---------------------------------------------|-------------------------------------|-----------------------|-------------------------------------|-----------------------|------|-----|---------------|-----|----------------|-----|-------|
|                                             |                                     |                       |                                     | MIN                   | TYP  | MAX | MIN           | MAX | MIN            | MAX |       |
| HC TYPES                                    |                                     |                       |                                     |                       |      |     |               |     |                |     |       |
| Propagation Delay<br>SH <sub>CP</sub> to Q7 | t <sub>PLH</sub> , t <sub>PHL</sub> | C <sub>L</sub> = 50pF | 2                                   | -                     | -    | 175 | -             | 220 | -              | 265 | ns    |
|                                             |                                     |                       | 4.5                                 | -                     | -    | 35  | -             | 44  | -              | 53  | ns    |
|                                             |                                     | C <sub>L</sub> =15pF  | 5                                   | -                     | 14   | -   | -             | -   | -              | -   | ns    |
|                                             |                                     | C <sub>L</sub> = 50pF | 6                                   | -                     | -    | 30  | -             | 37  | -              | 45  | ns    |
| P <sub>L</sub> to Q7                        | t <sub>PLH</sub> , t <sub>PHL</sub> | C <sub>L</sub> = 50pF | 2                                   | -                     | -    | 200 | -             | 250 | -              | 300 | ns    |
|                                             |                                     |                       | 4.5                                 | -                     | -    | 40  | -             | 50  | -              | 60  | ns    |
|                                             |                                     | C <sub>L</sub> =15pF  | 5                                   | -                     | 17   | -   | -             | -   | -              | -   | ns    |
|                                             |                                     | C <sub>L</sub> = 50pF | 6                                   | -                     | -    | 34  | -             | 43  | -              | 51  | ns    |
| ST <sub>CP</sub> to Q7                      | t <sub>PLH</sub> , t <sub>PHL</sub> | C <sub>L</sub> = 50pF | 2                                   | -                     | -    | 240 | -             | 300 | -              | 360 | ns    |
|                                             |                                     |                       | 4.5                                 | -                     | -    | 48  | -             | 60  | -              | 72  | ns    |
|                                             |                                     | C <sub>L</sub> =15pF  | 5                                   | -                     | 20   | -   | -             | -   | -              | -   | ns    |
|                                             |                                     | C <sub>L</sub> = 50pF | 6                                   | -                     | -    | 41  | -             | 51  | -              | 61  | ns    |
| M <sub>R</sub> to Q7                        | t <sub>PLH</sub> , t <sub>PHL</sub> | C <sub>L</sub> = 50pF | 2                                   | -                     | -    | 175 | -             | 220 | -              | 265 | ns    |
|                                             |                                     |                       | 4.5                                 | -                     | -    | 35  | -             | 44  | -              | 53  | ns    |
|                                             |                                     | C <sub>L</sub> =15pF  | 5                                   | -                     | 14   | -   | -             | -   | -              | -   | ns    |
|                                             |                                     | C <sub>L</sub> = 50pF | 6                                   | -                     | -    | 30  | -             | 37  | -              | 45  | ns    |
| Output Transition Time                      | t <sub>TLH</sub> , t <sub>THL</sub> | C <sub>L</sub> = 50pF | 2                                   | -                     | -    | 75  | -             | 95  | -              | 110 | ns    |
|                                             |                                     |                       | 4.5                                 | -                     | -    | 15  | -             | 19  | -              | 22  | ns    |
|                                             |                                     |                       | 6                                   | -                     | -    | 13  | -             | 16  | -              | 19  | ns    |
| Input Capacitance                           | C <sub>I</sub>                      | C <sub>L</sub> = 50pF | -                                   | -                     | -    | 10  | -             | 10  | -              | 10  | pF    |
| Power Dissipation Capacitance, (Notes 3, 4) | C <sub>PD</sub>                     | -                     | 5                                   | -                     | 13.5 | -   | -             | -   | -              | -   | pF    |
| HCT                                         |                                     |                       |                                     |                       |      |     |               |     |                |     |       |
| Propagation Delay<br>SH <sub>CP</sub> to Q7 | t <sub>PLH</sub> , t <sub>PHL</sub> | C <sub>L</sub> = 50pF | 4.5                                 | -                     | -    | 38  | -             | 48  | -              | 57  | ns    |
|                                             |                                     | C <sub>L</sub> = 15pF | 5                                   | -                     | 16   | -   | -             | -   | -              | -   | ns    |
|                                             |                                     | P <sub>L</sub> to Q7  | t <sub>PLH</sub> , t <sub>PHL</sub> | C <sub>L</sub> = 50pF | 4.5  | -   | -             | 48  | -              | 60  | -     |
| C <sub>L</sub> = 15pF                       | 5                                   |                       |                                     | -                     | 20   | -   | -             | -   | -              | -   | ns    |
| ST <sub>CP</sub> to Q7                      | t <sub>PLH</sub> , t <sub>PHL</sub> | C <sub>L</sub> = 50pF | 4.5                                 | -                     | -    | 56  | -             | 70  | -              | 84  | ns    |
|                                             |                                     | C <sub>L</sub> = 15pF | 5                                   | -                     | 23   | -   | -             | -   | -              | -   | ns    |

# CD54HC597, CD74HC597, CD74HCT597

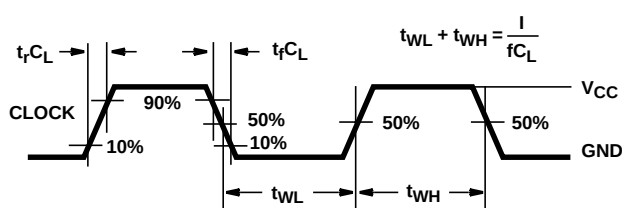
## Switching Specifications Input $t_r, t_f = 6\text{ns}$ (Continued)

| PARAMETER                                   | SYMBOL             | TEST CONDITIONS     | $V_{CC}$ (V) | 25°C |      |     | -40°C to 85°C |     | -55°C to 125°C |     | UNITS |
|---------------------------------------------|--------------------|---------------------|--------------|------|------|-----|---------------|-----|----------------|-----|-------|
|                                             |                    |                     |              | MIN  | TYP  | MAX | MIN           | MAX | MIN            | MAX |       |
| $\overline{MR}$ to Q7                       | $t_{PLH}, t_{PHL}$ | $C_L = 50\text{pF}$ | 4.5          | -    | -    | 44  | -             | 55  | -              | 66  | ns    |
|                                             |                    | $C_L = 15\text{pF}$ | 5            | -    | 18   | -   | -             | -   | -              | -   | ns    |
| Output Transition Time                      | $t_{TLH}, t_{THL}$ | $C_L = 50\text{pF}$ | 4.5          | -    | -    | 15  | -             | 19  | -              | 22  | ns    |
| Input Capacitance                           | $C_I$              | $C_L = 50\text{pF}$ | -            | -    | -    | 10  | -             | 10  | -              | 10  | pF    |
| Power Dissipation Capacitance, (Notes 3, 4) | $C_{PD}$           | -                   | 5            | -    | 18.5 | -   | -             | -   | -              | -   | pF    |

### NOTES:

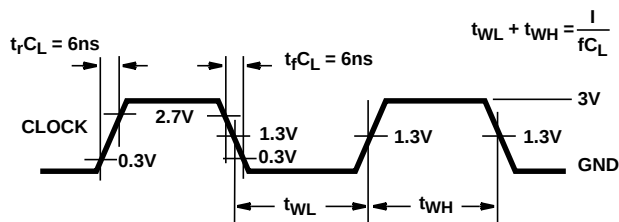
- $C_{PD}$  is used to determine the dynamic power consumption, per package.
- $P_D = C_{PD} V_{CC}^2 f_i + \sum (C_L V_{CC}^2 f_o)$  where:  $f_i$  = Input Frequency,  $f_o$  = Output Frequency,  $C_L$  = Output Load Capacitance,  $V_{CC}$  = Supply Voltage.

## Test Circuits and Waveforms



NOTE: Outputs should be switching from 10%  $V_{CC}$  to 90%  $V_{CC}$  in accordance with device truth table. For  $f_{MAX}$ , input duty cycle = 50%.

FIGURE 1. HC CLOCK PULSE RISE AND FALL TIMES AND PULSE WIDTH



NOTE: Outputs should be switching from 10%  $V_{CC}$  to 90%  $V_{CC}$  in accordance with device truth table. For  $f_{MAX}$ , input duty cycle = 50%.

FIGURE 2. HCT CLOCK PULSE RISE AND FALL TIMES AND PULSE WIDTH

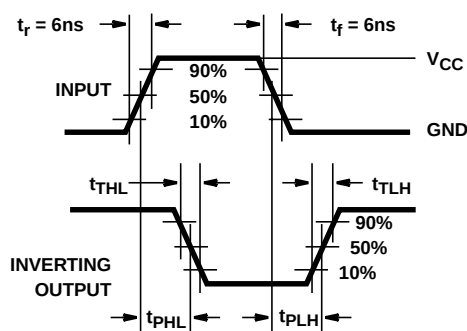


FIGURE 3. HC TRANSITION TIMES AND PROPAGATION DELAY TIMES, COMBINATION LOGIC

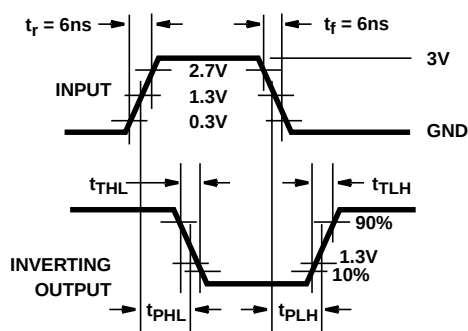


FIGURE 4. HCT TRANSITION TIMES AND PROPAGATION DELAY TIMES, COMBINATION LOGIC

**Test Circuits and Waveforms** (Continued)

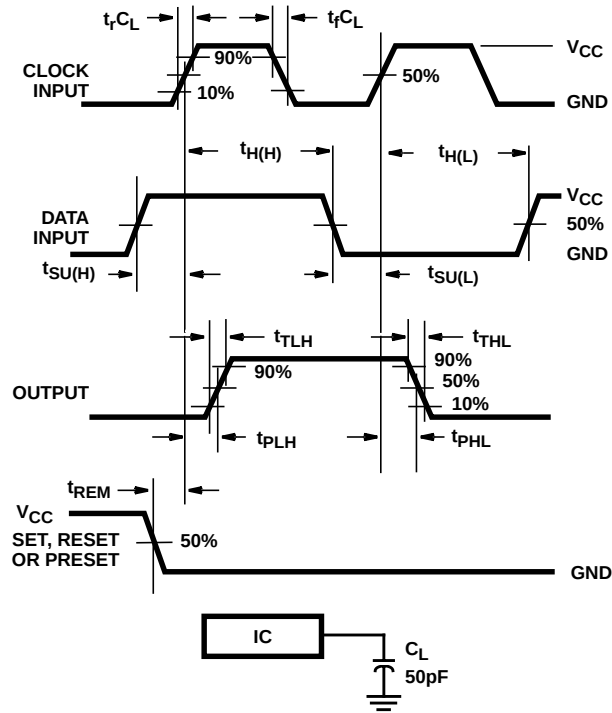


FIGURE 5. HC SETUP TIMES, HOLD TIMES, REMOVAL TIME, AND PROPAGATION DELAY TIMES FOR EDGE TRIGGERED SEQUENTIAL LOGIC CIRCUITS

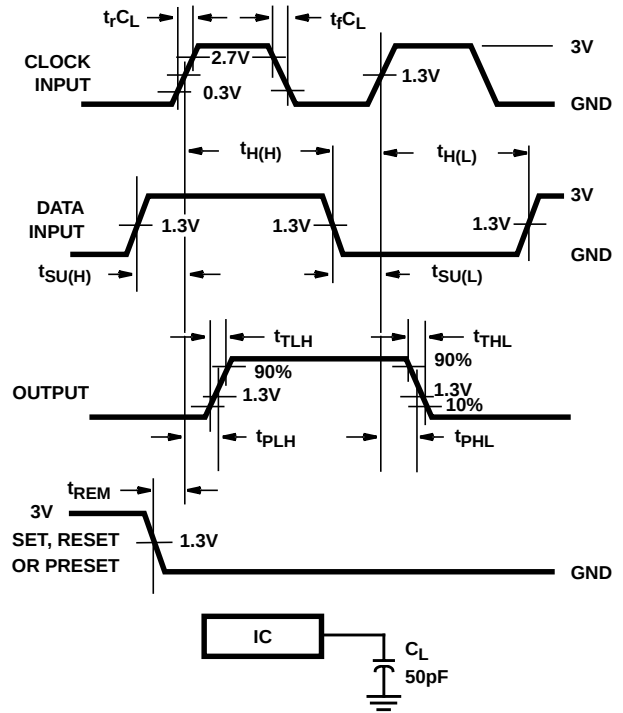
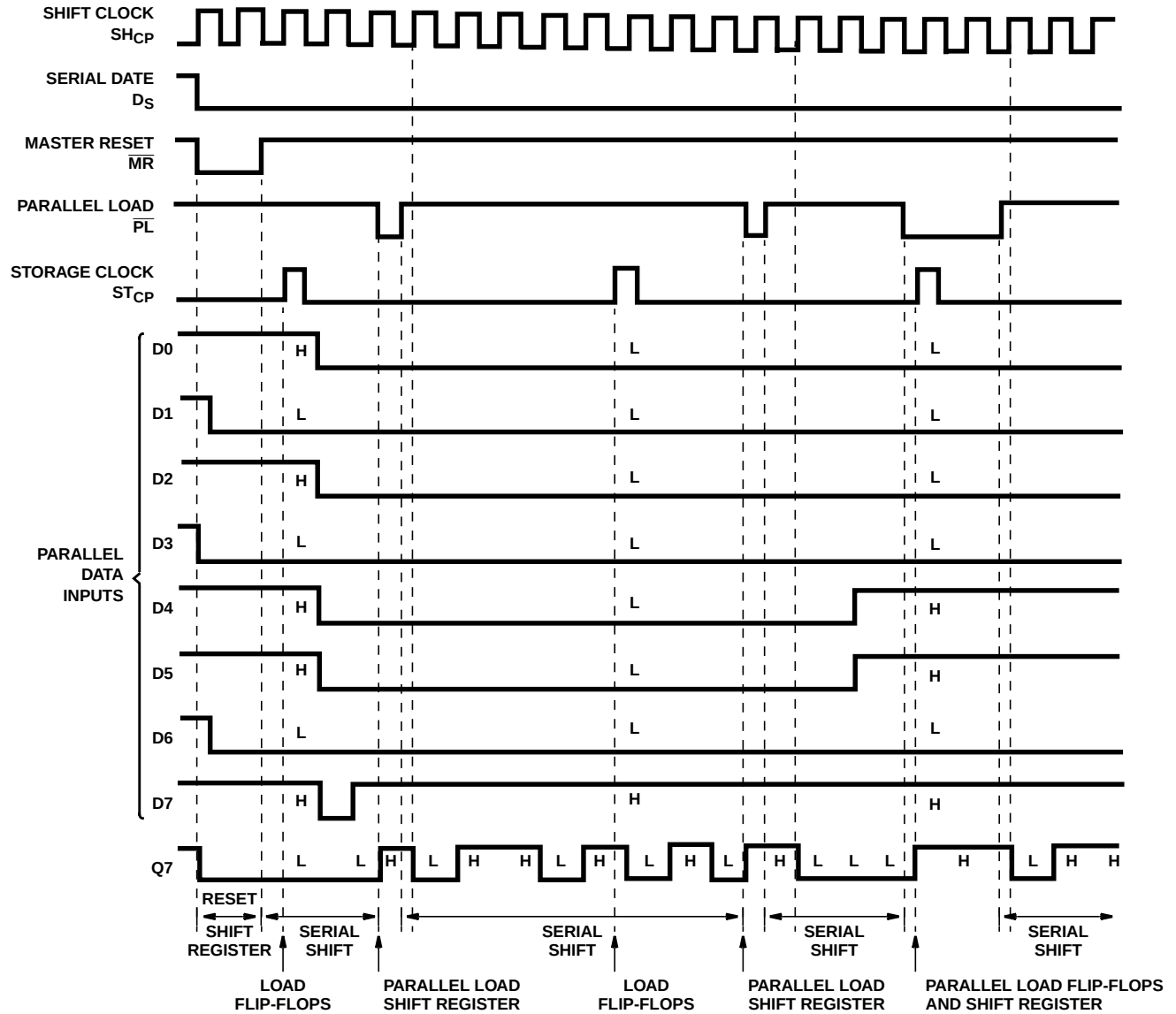


FIGURE 6. HCT SETUP TIMES, HOLD TIMES, REMOVAL TIME, AND PROPAGATION DELAY TIMES FOR EDGE TRIGGERED SEQUENTIAL LOGIC CIRCUITS

# Timing Diagram



## PACKAGING INFORMATION

| Orderable part number          | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)            |
|--------------------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|--------------------------------|
| <a href="#">5962-8681701EA</a> | Active        | Production           | CDIP (J)   16  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 5962-8681701EA<br>CD54HC597F3A |
| <a href="#">CD54HC597F3A</a>   | Active        | Production           | CDIP (J)   16  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 5962-8681701EA<br>CD54HC597F3A |
| CD54HC597F3A.A                 | Active        | Production           | CDIP (J)   16  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 5962-8681701EA<br>CD54HC597F3A |
| <a href="#">CD74HC597E</a>     | Active        | Production           | PDIP (N)   16  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -55 to 125   | CD74HC597E                     |
| CD74HC597E.A                   | Active        | Production           | PDIP (N)   16  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -55 to 125   | CD74HC597E                     |
| CD74HC597EE4                   | Active        | Production           | PDIP (N)   16  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -55 to 125   | CD74HC597E                     |
| <a href="#">CD74HC597M</a>     | Obsolete      | Production           | SOIC (D)   16  | -                     | -           | Call TI                              | Call TI                           | -55 to 125   | HC597M                         |
| <a href="#">CD74HC597M96</a>   | Active        | Production           | SOIC (D)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HC597M                         |
| CD74HC597M96.A                 | Active        | Production           | SOIC (D)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HC597M                         |
| CD74HC597M96E4                 | Active        | Production           | SOIC (D)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HC597M                         |
| CD74HC597M96G4                 | Active        | Production           | SOIC (D)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HC597M                         |
| <a href="#">CD74HC597MT</a>    | Obsolete      | Production           | SOIC (D)   16  | -                     | -           | Call TI                              | Call TI                           | -55 to 125   | HC597M                         |
| <a href="#">CD74HC597NSR</a>   | Active        | Production           | SOP (NS)   16  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HC597M                         |
| CD74HC597NSR.A                 | Active        | Production           | SOP (NS)   16  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HC597M                         |
| <a href="#">CD74HCT597E</a>    | Active        | Production           | PDIP (N)   16  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -55 to 125   | CD74HCT597E                    |
| CD74HCT597E.A                  | Active        | Production           | PDIP (N)   16  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -55 to 125   | CD74HCT597E                    |
| <a href="#">CD74HCT597M</a>    | Obsolete      | Production           | SOIC (D)   16  | -                     | -           | Call TI                              | Call TI                           | -55 to 125   | HCT597M                        |
| <a href="#">CD74HCT597M96</a>  | Active        | Production           | SOIC (D)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HCT597M                        |
| CD74HCT597M96.A                | Active        | Production           | SOIC (D)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -55 to 125   | HCT597M                        |
| <a href="#">CD74HCT597MT</a>   | Obsolete      | Production           | SOIC (D)   16  | -                     | -           | Call TI                              | Call TI                           | -55 to 125   | HCT597M                        |

<sup>(1)</sup> **Status:** For more details on status, see our [product life cycle](#).

<sup>(2)</sup> **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

<sup>(3)</sup> **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

**(4) Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

**(5) MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

**(6) Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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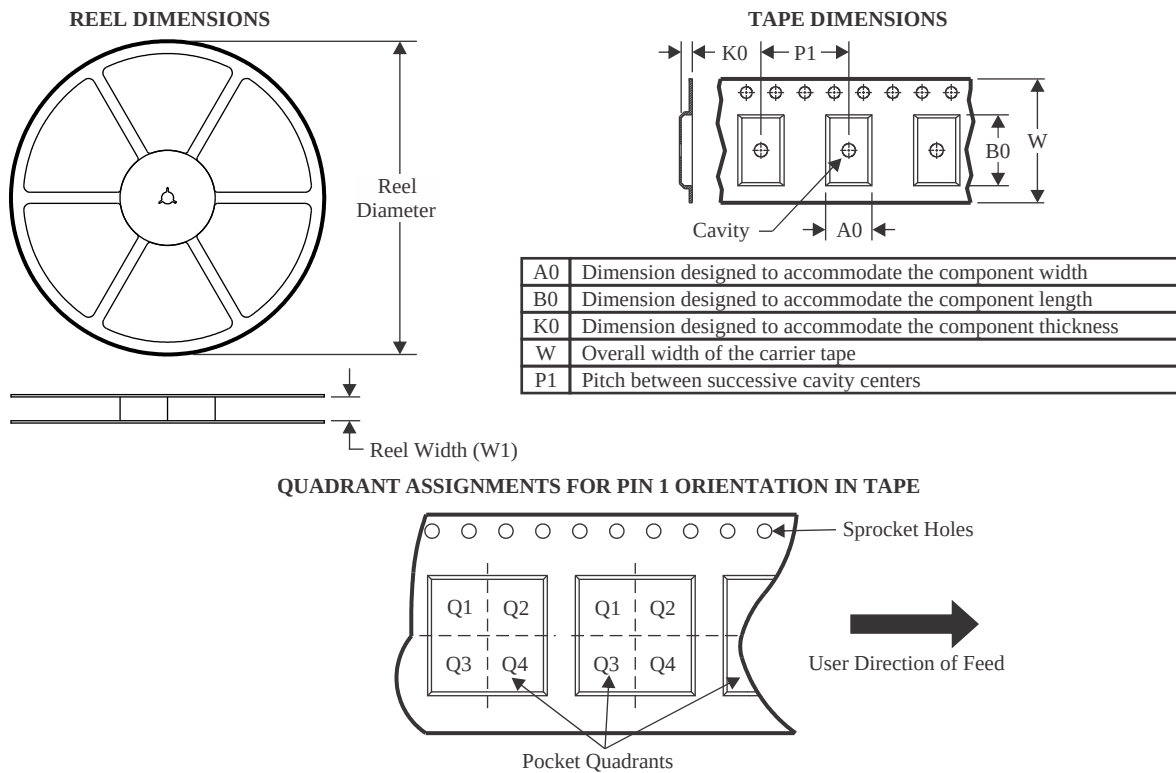
**OTHER QUALIFIED VERSIONS OF CD54HC597, CD74HC597 :**

- Catalog : [CD74HC597](#)
- Military : [CD54HC597](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

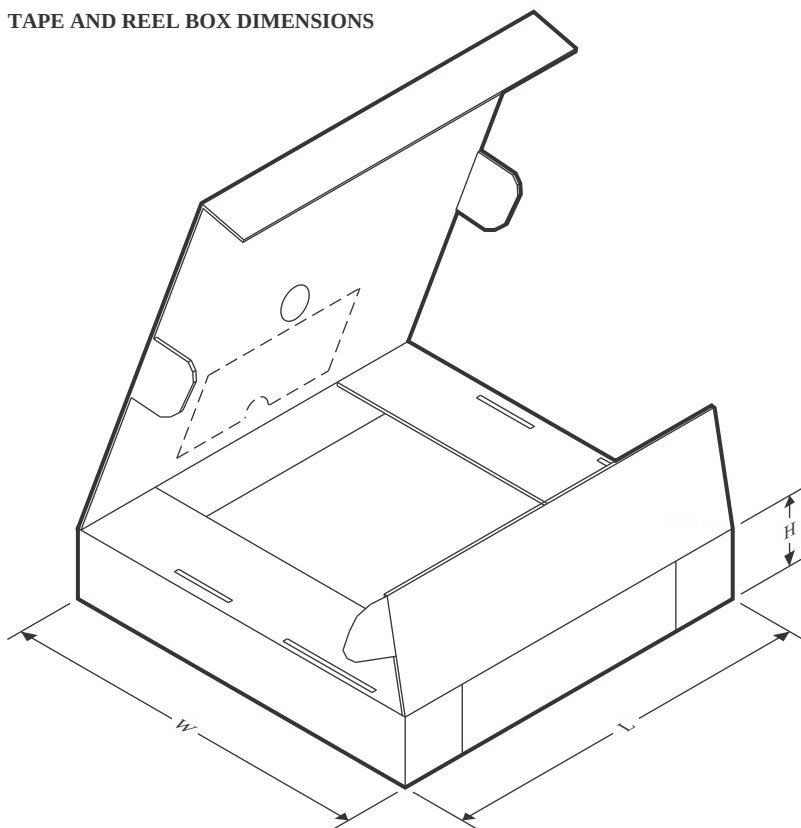
## TAPE AND REEL INFORMATION



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| CD74HC597M96  | SOIC         | D               | 16   | 2500 | 330.0              | 16.4               | 6.5     | 10.3    | 2.1     | 8.0     | 16.0   | Q1            |
| CD74HC597NSR  | SOP          | NS              | 16   | 2000 | 330.0              | 16.4               | 8.1     | 10.4    | 2.5     | 12.0    | 16.0   | Q1            |
| CD74HCT597M96 | SOIC         | D               | 16   | 2500 | 330.0              | 16.4               | 6.5     | 10.3    | 2.1     | 8.0     | 16.0   | Q1            |

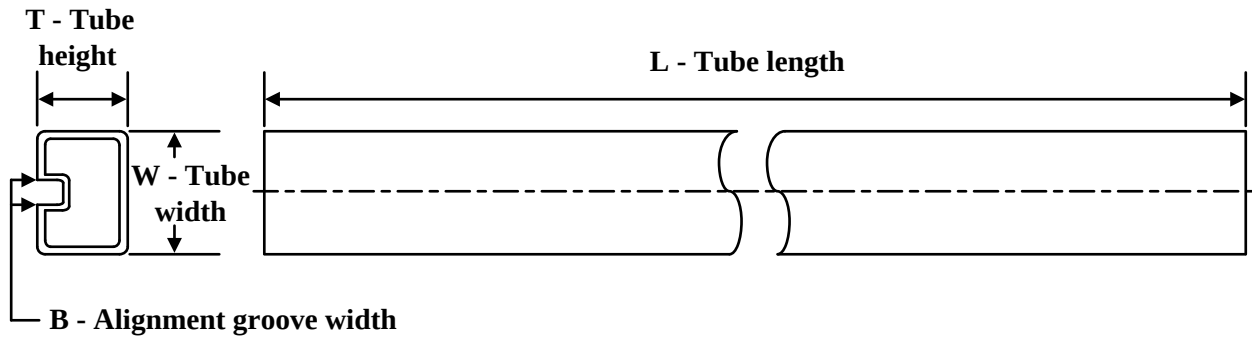
## TAPE AND REEL BOX DIMENSIONS



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|---------------|--------------|-----------------|------|------|-------------|------------|-------------|
| CD74HC597M96  | SOIC         | D               | 16   | 2500 | 353.0       | 353.0      | 32.0        |
| CD74HC597NSR  | SOP          | NS              | 16   | 2000 | 353.0       | 353.0      | 32.0        |
| CD74HCT597M96 | SOIC         | D               | 16   | 2500 | 353.0       | 353.0      | 32.0        |

## TUBE

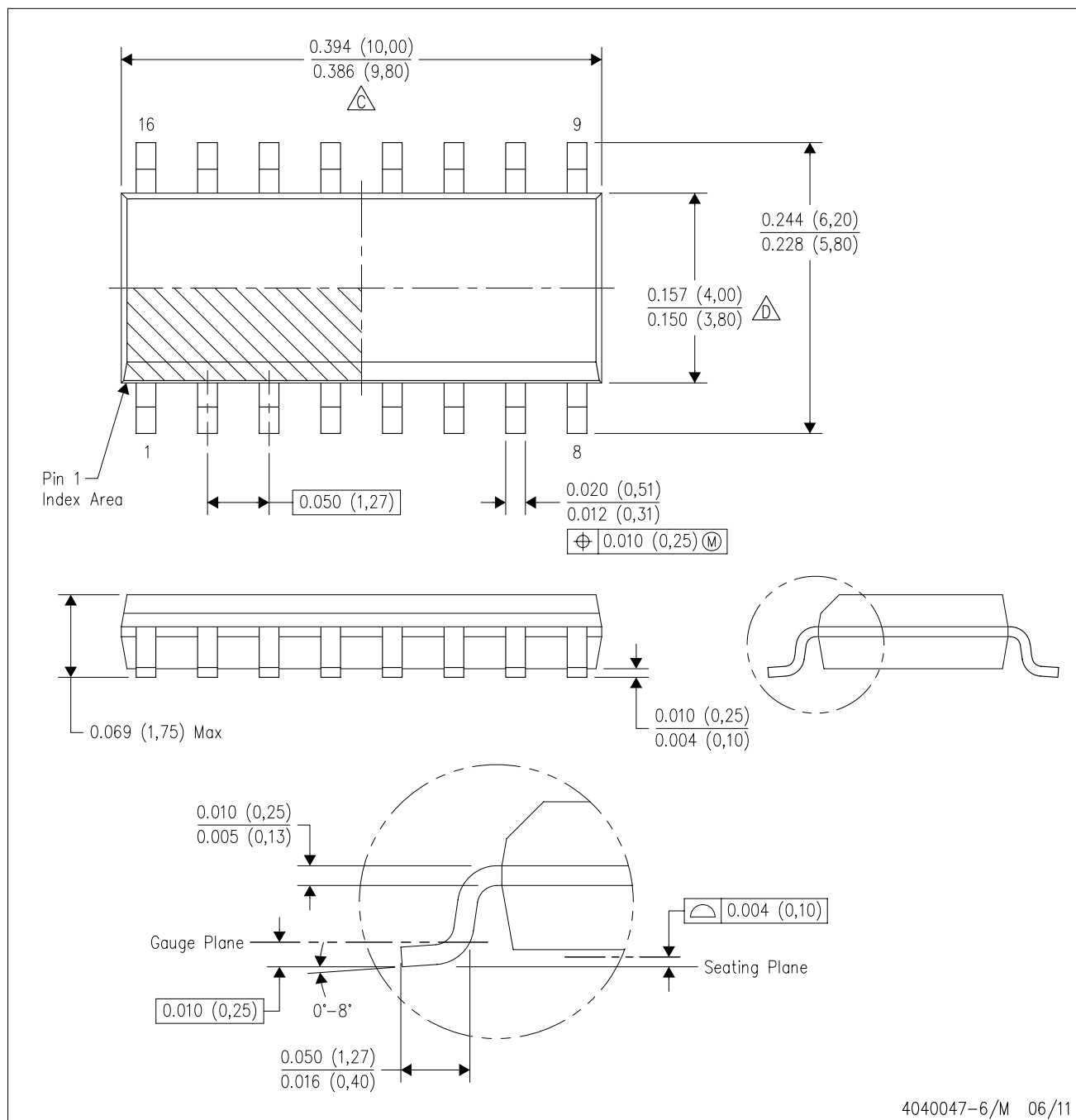


\*All dimensions are nominal

| Device        | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|---------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| CD74HC597E    | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HC597E    | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HC597E.A  | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HC597E.A  | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HC597EE4  | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HC597EE4  | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HCT597E   | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HCT597E   | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HCT597E.A | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |
| CD74HCT597E.A | N            | PDIP         | 16   | 25  | 506    | 13.97  | 11230  | 4.32   |

D (R-PDSO-G16)

PLASTIC SMALL OUTLINE



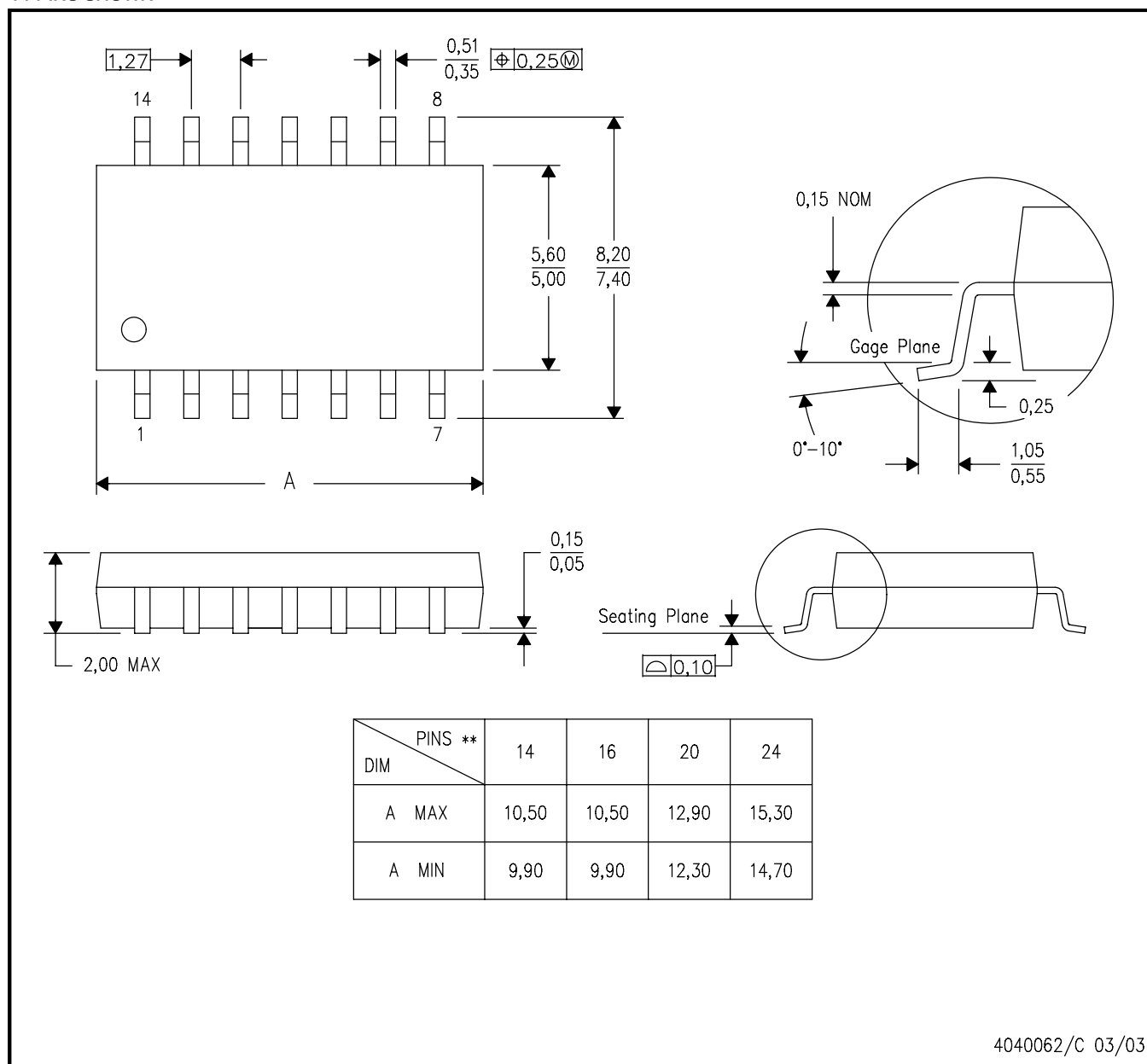
- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  - $\triangle C$  Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
  - $\triangle D$  Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
  - E. Reference JEDEC MS-012 variation AC.

# MECHANICAL DATA

NS (R-PDSO-G\*\*)

PLASTIC SMALL-OUTLINE PACKAGE

14-PINS SHOWN

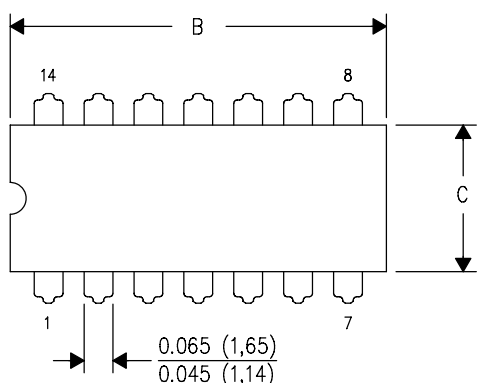


- NOTES:
- All linear dimensions are in millimeters.
  - This drawing is subject to change without notice.
  - Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

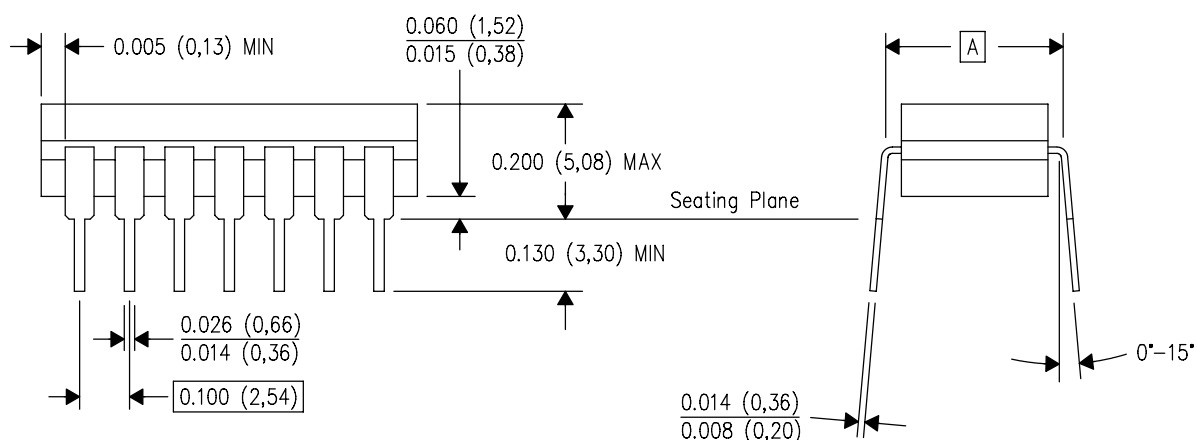
J (R-GDIP-T\*\*)

14 LEADS SHOWN

# CERAMIC DUAL IN-LINE PACKAGE



| PINS **<br>DIM | 14                     | 16                     | 18                     | 20                     |
|----------------|------------------------|------------------------|------------------------|------------------------|
| A              | 0.300<br>(7,62)<br>BSC | 0.300<br>(7,62)<br>BSC | 0.300<br>(7,62)<br>BSC | 0.300<br>(7,62)<br>BSC |
| B MAX          | 0.785<br>(19,94)       | .840<br>(21,34)        | 0.960<br>(24,38)       | 1.060<br>(26,92)       |
| B MIN          | —                      | —                      | —                      | —                      |
| C MAX          | 0.300<br>(7,62)        | 0.300<br>(7,62)        | 0.310<br>(7,87)        | 0.300<br>(7,62)        |
| C MIN          | 0.245<br>(6,22)        | 0.245<br>(6,22)        | 0.220<br>(5,59)        | 0.245<br>(6,22)        |



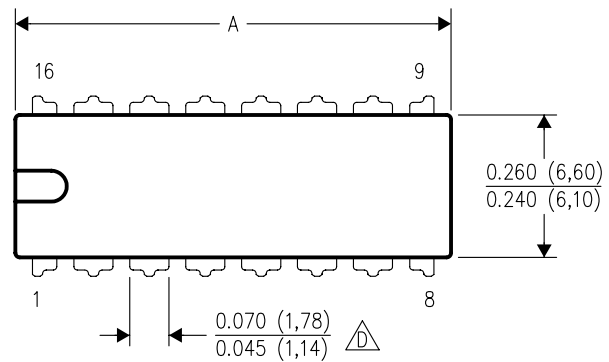
4040083/F 03/03

- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  - C. This package is hermetically sealed with a ceramic lid using glass frit.
  - D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
  - E. Falls within MIL STD 1835 GDIP1-T14, GDIP1-T16, GDIP1-T18 and GDIP1-T20.

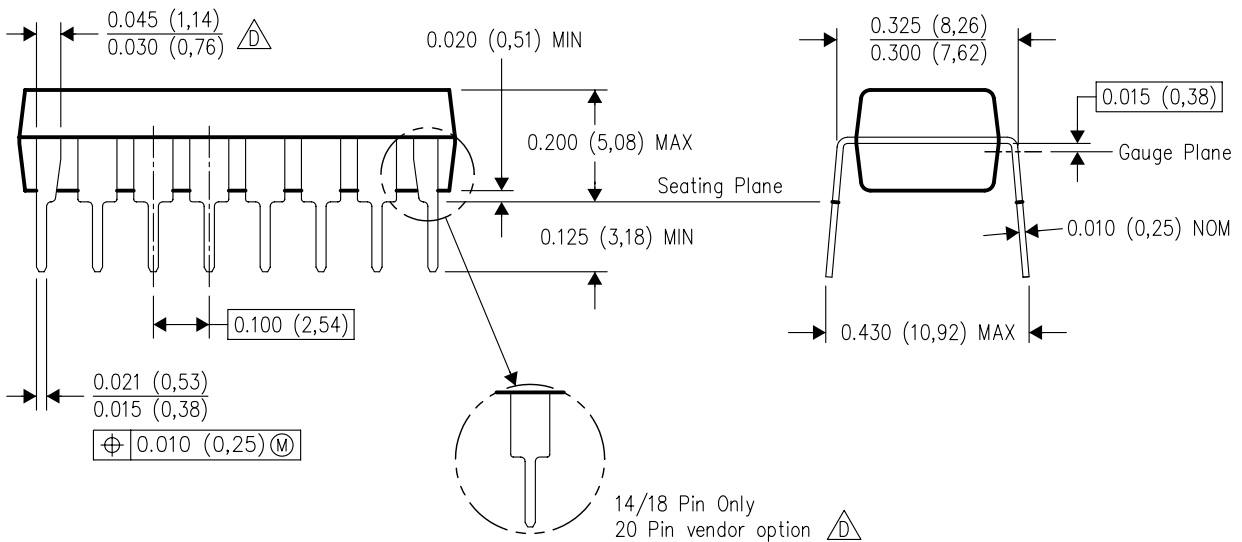
## N (R-PDIP-T\*\*)

16 PINS SHOWN

## PLASTIC DUAL-IN-LINE PACKAGE



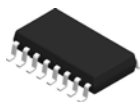
| PINS **             | 14               | 16               | 18               | 20               |
|---------------------|------------------|------------------|------------------|------------------|
| DIM                 |                  |                  |                  |                  |
| A MAX               | 0.775<br>(19,69) | 0.775<br>(19,69) | 0.920<br>(23,37) | 1.060<br>(26,92) |
| A MIN               | 0.745<br>(18,92) | 0.745<br>(18,92) | 0.850<br>(21,59) | 0.940<br>(23,88) |
| MS-001<br>VARIATION | AA               | BB               | AC               | AD               |



14/18 Pin Only  
20 Pin vendor option

4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  - Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
  - The 20 pin end lead shoulder width is a vendor option, either half or full width.

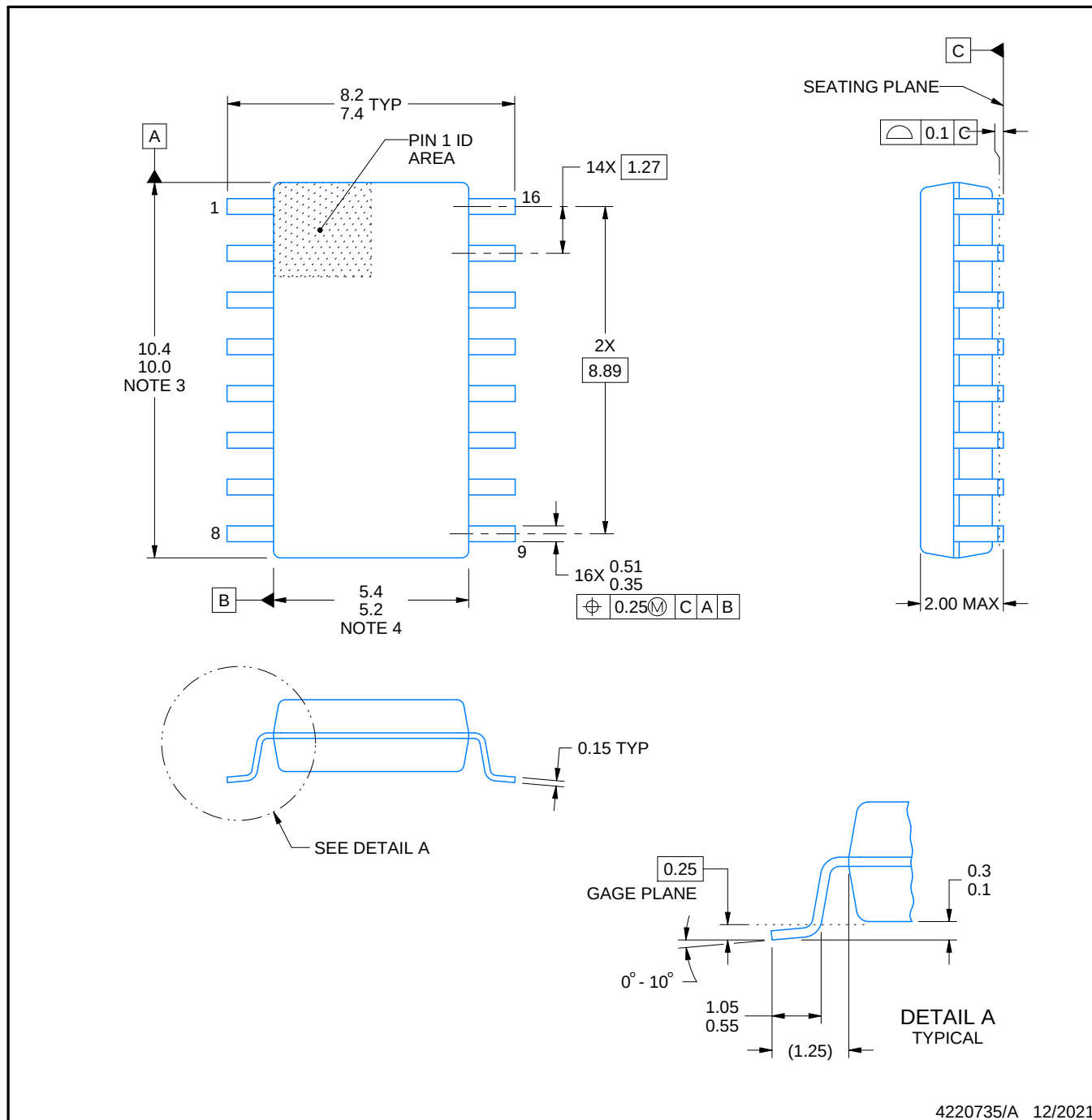


# PACKAGE OUTLINE

NS0016A

SOP - 2.00 mm max height

SOP

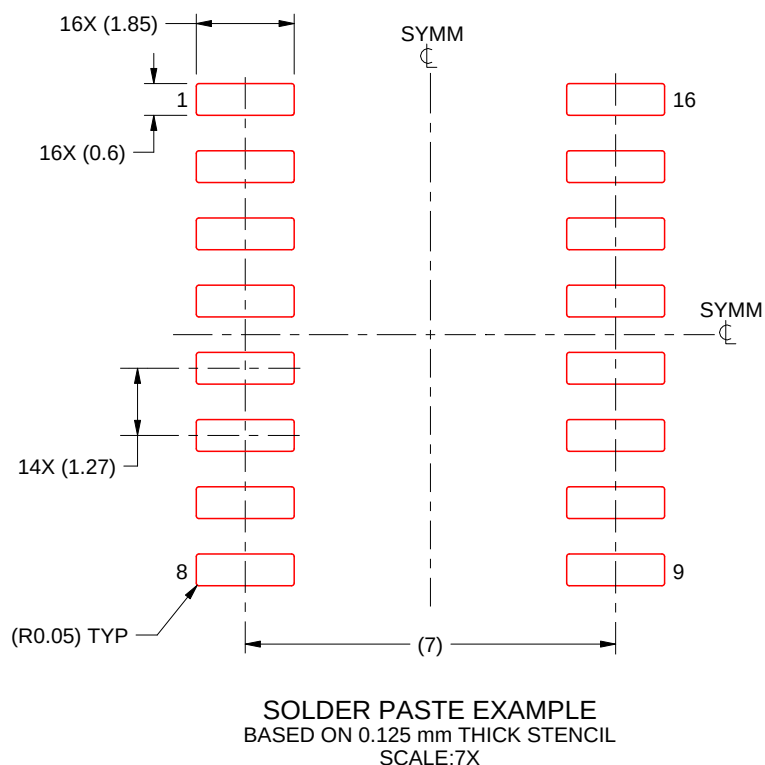


4220735/A 12/2021

## NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.





4220735/A 12/2021

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

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